

JORGE J. ORTIZ

AI Evaluation & Research Leader | Trustworthy Agentic Systems

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Research Leadership Profile

Hands-on AI research leader who builds evaluation systems for models and agents operating under real-world uncertainty. Research connects long-horizon agent evaluation, formal verification, runtime monitoring, multimodal capability measurement, and explicit abstention. The central question is not only what a system can do, but when its evidence is sufficient to act. Director of the Sensing & Reasoning Lab; Rutgers site director or lead for two NSF centers; Associate Professor with tenure; and applied AI lead in professional sports. Experienced setting research direction, mentoring researchers and engineers, writing production-oriented research code, and communicating limitations and evidence to technical and non-technical stakeholders.

Relevant Research & Evaluation Leadership

Director, Sensing & Reasoning Lab

Rutgers University

2018–Present

Piscataway, NJ

- ▶ Set the research agenda for trustworthy agentic AI, multimodal reasoning, causal inference, and self-assessing systems; lead projects from hypothesis and experimental design through implementation, deployment, publication, and external communication.
- ▶ Mentor multidisciplinary researchers and engineers; graduated three Ph.D. students and lead collaborations spanning computer science, engineering, human–robot interaction, health, and urban systems.
- ▶ Build evaluation methods around explicit competency boundaries, calibrated uncertainty, failure analysis, and human escalation instead of benchmark score alone.

Rutgers Site Director, Center for Responsible AI & Governance (CRAIG)

NSF Industry–University Cooperative Research Center

2025–Present

Rutgers University

- ▶ Lead a cross-sector research program on the evidence required to justify AI deployment, translating technical evaluation results into reusable governance and decision frameworks.
- ▶ Develop evaluation pipelines for agentic systems that combine formal guarantees, runtime evidence, guardrails, and auditable post-training behavior.

Rutgers Site Lead, Center for Smart Streetscapes (CS3)

NSF Engineering Research Center

2022–Present

Rutgers University

- ▶ Lead Rutgers research within a \$26M multi-institution center; align sensing, perception, privacy, and causal reasoning work across academic, government, community, and industry partners.
- ▶ Drive evaluation in deployed settings where distribution shift, incomplete evidence, and operational constraints make capability boundaries consequential.

Selected Work in Model Capabilities, Agents & Measurement

Veritas for Epistemic Oversight of Long-Running AI Agents

Research system and multi-model evaluation study under preparation for submission

2026–Present

- ▶ Veritas tests whether long-running AI agents can support their claims with evidence and recognize when they should proceed, revise their work, or defer to a human.
- ▶ The system turns claims into explicit verification obligations, runs independent checks, and preserves auditable traces and evidence bundles for each decision.
- ▶ My group and I are preparing this work for submission. The study uses matched conditions across multiple model families, seeded faults, clean-task do-no-harm panels, independent scoring, and predeclared budgets and stopping rules.

TraceFix for Verified Coordination in Agentic Systems

Outstanding Solution Paper

ACM CAIS 2026

- ▶ Use TLA+ model checking and counterexample-guided repair to find and fix deadlocks, races, and protocol violations in multi-agent coordination; enforce verified protocols with runtime topology monitoring.

Explicit Abstention for Predictable VLM Reliability

Preprint

2026

- ▶ Developed explicit abstention controls for video question answering to expose capability–coverage tradeoffs and make system reliability more predictable under uncertainty.

Human-Centered RLHF for Autonomous Driving

Model behavior and safety

IMWUT/UbiComp 2024

- ▶ Co-developed an LLM-enhanced RLHF approach that incorporates human-centered safety preferences into autonomous-driving optimization and evaluation.

Professional Experience

Associate Professor (with Tenure), Electrical & Computer Engineering <i>Rutgers University</i>	2025–Present <i>Piscataway, NJ</i>
▶ Principal investigator or co-investigator on a \$32.4M portfolio across responsible AI, urban AI, health sensing, and intelligent infrastructure; balance scientific rigor with milestones, partner needs, and deployed-system constraints. ▶ Leadership Research Faculty Fellow (2026), developing best practices for thoughtful AI use in research development.	
Assistant Professor, Electrical & Computer Engineering <i>Rutgers University</i>	2018–2025 <i>Piscataway, NJ</i>
AI & Computer Vision Lead, Baseball Operations <i>New York Yankees</i>	2019–Present <i>New York, NY</i>
▶ Lead applied computer-vision and biomechanics research for player development and performance, with an emphasis on robust measurement and signaling uncertainty in high-stakes decisions.	
Research Staff Member <i>IBM Research</i>	2013–2018 <i>Yorktown Heights, NY</i>
▶ Researched and built machine-learning, sensing, and distributed systems that moved from experimental prototypes toward real operating environments.	
Senior Software Engineer / Software Engineer <i>Spire Global / Oracle</i>	2003–2013 <i>San Francisco Bay Area</i>

Selected Publications & Research Signals

- ▶ **Xia, Li, Ehsan, Ortiz.** “TraceFix, Repairing Agent Coordination Protocols with TLA+ Counterexamples.” *ACM Conference on AI and Society*, 2026. Outstanding Solution Paper.
- ▶ **Ortiz.** “Explicit Abstention Knobs for Predictable Reliability in Video Question Answering.” arXiv 2601.00138, 2026.
- ▶ **Sun, Contreras, Ortiz.** “DFGauss, Dynamic Focused Masking for Autoregressive 3D Occupancy Prediction.” *NeurIPS*, 2025.
- ▶ **Bu, Tsai, Tjokro, Bhattacharjee, Ortiz, Ju.** “Using Vision-Language Models as Proxies for Social Intelligence in Human–Robot Interaction.” 2025.
- ▶ Research has appeared across NeurIPS, CoRL, IMWUT/UbiComp, IPSN, SenSys, IoTDI, and USENIX ATC; recipient of multiple best-paper and best-paper-finalist recognitions.

Technical & Organizational Strengths

Evaluation	Long-horizon agents; capability and coverage measurement; fault injection; matched experiments; independent scoring; ablations; negative-result analysis; reproducible evidence
Trustworthy AI	Formal methods (TLA+); runtime monitoring; verification plans; evidence provenance; abstention; uncertainty calibration; causal reasoning; privacy-aware sensing
Research leadership	Strategy and roadmaps; team mentorship; cross-functional collaboration; multi-institution programs; model-release-style technical narratives; stakeholder communication
Implementation	Python, Go, C/C++; evaluation harnesses; ML/CV systems; distributed systems; experiment automation; open-source research artifacts

Education

Ph.D., Computer Science , University of California, Berkeley	2013
M.S., Computer Science , University of California, Berkeley	2010
B.S., Computer Science , Massachusetts Institute of Technology	2003